

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

COOLING TOWER SYSTEM

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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1. System Profiling & Overview

A cooling tower rejects waste heat from process/HVAC systems to the atmosphere primarily through evaporative cooling. Warm water from condensers/heat exchangers is distributed over fill media while ambient air is drawn/forced through it, cooling the water via evaporation and sensible heat transfer before it is recirculated back to the process.

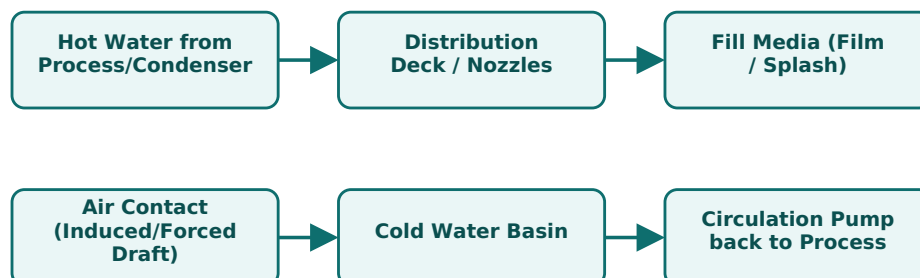


Fig. 1 — Cooling Tower Water & Air Contact Path (counter-flow induced-draft example)

1.1 Applications

- Chiller/HVAC condenser water cooling for commercial buildings and data centers.
- Process cooling for power plants, refineries, petrochemical and steel plants.
- Compressor and reactor jacket cooling in chemical/pharma manufacturing.
- Injection molding and other manufacturing process cooling.

1.2 Key Performance Terms

- **Range:** Difference between hot water inlet and cold water outlet temperature.
- **Approach:** Difference between cold water outlet temperature and ambient wet bulb temperature (lower approach = larger/more efficient tower).
- **Cycles of Concentration (COC):** Ratio of dissolved solids in circulating water to make-up water, controlled via blowdown; typically 3-6 cycles.
- **Drift Loss:** Water droplets carried out with exhaust air, minimized (<0.005% of recirculation rate) by drift eliminators.
- **Evaporation Loss:** Approximately 1% of circulation rate per 5.5°C range.

2. Types of Cooling Towers & Selection Basis

Type	Description	Typical Application
Natural Draft (Hyperbolic)	Uses buoyancy/stack effect, no fans, very large size	Large power plants (500+ MW)
Induced Draft Counter-flow	Fan at top pulls air upward against falling water	Industrial process cooling, refineries
Induced Draft Cross-flow	Air moves horizontally across falling water	Large HVAC/industrial installations
Forced Draft	Fan at base pushes air through tower	Compact/low-noise installations
FRP Modular (Induced Draft)	Factory-built FRP cells, field-assembled	Commercial buildings, mid-size industrial plants
Closed-circuit (Fluid Cooler)	Process fluid in closed coil, sprayed externally, no direct contact	Where process fluid purity must be protected from contamination

3. Thermal & Process Design

- Heat load (kcal/hr or TR) determines required water circulation rate: $Q = m \times C_p \times \text{Range}$.
- Design wet bulb temperature (site-specific meteorological data) governs achievable cold water temperature; approach of 3-5°C above design WBT is common for economical design.
- L/G ratio (liquid-to-gas mass flow ratio) and fill selection (film fill vs splash fill) determine tower fill volume and fan

horsepower.

- Fill selection: film fill (high efficiency, needs cleaner water) vs splash fill (more fouling tolerant, used where water quality is poor or effluent-laden).
- Fan and motor sizing based on air flow requirement with 10-15% safety margin; VFD control recommended for part-load energy savings.
- Basin sizing to hold sufficient water inventory considering pump NPSH and system holdup volume.

4. Make-up Water Pretreatment

Pretreatment Stage	Purpose	Typical Equipment
Multimedia/Sand Filtration	Remove suspended solids/turbidity	Pressure sand filter
Softening	Reduce hardness causing scale (see Softener manual)	Ion-exchange softener
Side-stream Filtration	Continuously polish circulating water	Sand or automatic self-cleaning screen filter on a 5-10% side-stream
Chemical Dosing System	Scale/corrosion/biological control	Dosing pumps for scale inhibitor, corrosion inhibitor, biocide

5. Water Chemistry Management

- **Scale Control:** Scale/antiscalant dosing (phosphonates, polymers) to keep calcium carbonate and silica in solution; blowdown control to limit COC below the Langelier Saturation Index (LSI) threshold for the specific water chemistry.
- **Corrosion Control:** Corrosion inhibitors (molybdate, zinc, or all-organic blends), pH control (target 7.0-8.5), monitored via corrosion coupons/corrometer readings.
- **Biological (Microbiological) Control:** Oxidizing biocides (chlorine/bromine-based) and non-oxidizing biocides (isothiazolone, glutaraldehyde) dosed on shock/continuous basis to control algae, biofilm and Legionella risk.
- **Blowdown Control:** Conductivity-based automatic blowdown valve maintains target COC, balancing water conservation against scaling risk.

6. Cooling Water Chemistry Deep-Dive

6.1 Langelier Saturation Index (LSI) & Scaling Tendency

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LSI = pH(actual) - pHs
pHs = (9.3 + A + B) - (C + D)
where A = f(TDS), B = f(temperature), C = f(calcium hardness), D = f(alkalinity)

LSI > 0 → scaling tendency (CaCO3 will tend to precipitate)
LSI = 0 → water in calcium carbonate equilibrium
LSI < 0 → water is corrosive / under-saturated, scale will dissolve
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Operating COC is normally set so that circulating-water LSI stays at or below the value the antiscalant program is qualified for (commonly LSI 2.0-2.5 with a good phosphonate/polymer program, vs LSI ≤ 1.0 -1.5 with no chemical treatment). Raising COC beyond this point without adjusting the antiscalant dose is the single most common self-inflicted cause of scaling.

6.2 Cycles of Concentration & Mass Balance

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COC = TDS(circulating) / TDS(make-up) ≈ (E + B + D) / (B + D)
where E = Evaporation, B = Blowdown, D = Drift, all in the same flow units

Worked example – 1000 m3/hr circulation, 1.2% evaporation, target COC = 4:
Evaporation (E) = 12 m3/hr
Required blowdown (B) = E / (COC-1) = 12 / 3 = 4 m3/hr
Make-up (M) = E + B + Drift ≈ 12 + 4 + 0.05 = 16.05 m3/hr
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This mass balance is the basis for sizing the make-up water source, softener/side-stream filter, and the blowdown control valve; it is also the calculation used to verify a site's actual COC against its design target during troubleshooting.

6.3 Corrosion Chemistry & Inhibitor Mechanisms

Uninhibited mild steel corrodes via the oxygen-reduction/iron-oxidation couple ($2\text{Fe} + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Fe}(\text{OH})_2$, further oxidizing to rust). Molybdate and orthophosphate inhibitors work as anodic passivators, forming a protective oxide film on the metal surface; zinc works as a cathodic inhibitor, precipitating as zinc hydroxide at cathodic sites. Corrosion rate is tracked quantitatively via corrosion coupons (weight-loss method, expressed in mils-per-year, mpy) – a rate below 3-5 mpy on mild steel is a typical acceptance criterion for a well-controlled system.

6.4 Biocide Chemistry & Legionella Control

Oxidizing biocides (chlorine/bromine) act by disrupting microbial cell membranes and enzymes; their effectiveness is governed by CT (concentration $\times$ contact time) rather than concentration alone. Non-oxidizing biocides (isothiazolone, glutaraldehyde) attack specific cellular targets and are typically alternated with oxidizing biocides to prevent microbial resistance. Legionella risk is highest in the 20-45°C range typical of cooling water – routine free-residual monitoring (0.5-1 ppm free chlorine equivalent) combined with periodic shock dosing and biofilm-focused non-oxidizing biocide rotation is the standard control strategy.

6.5 Silica Scaling Limit

Unlike calcium carbonate, silica scaling is not pH-driven and cannot be controlled by acid dosing – it is purely a solubility limit (amorphous silica solubility is ~ 100 -150 mg/L at typical cooling water temperature and pH). Once circulating-water silica (feed silica $\times$ COC) approaches this limit, further COC increase must stop regardless of the calcium carbonate LSI, since silica scale is hard, glassy and very difficult to remove chemically once deposited.

7. Common Problems & Engineering Solutions

Problem	Root Cause	Engineering Solution
Scaling on fill/condenser tubes	High COC, poor antiscalant dosing, hard make-up water	Soften make-up water, optimize blowdown, verify antiscalant dose via LSI monitoring
Corrosion of pipework/heat	Low pH, oxygen ingress, inadequate inhibitor	Maintain corrosion inhibitor residual, monitor via coupons,

exchangers	dose	control pH
Algae growth in basin/fill	Sunlight exposure, insufficient biocide dosing	Cover basin, increase biocide dosing frequency, consider UV/ozone treatment
Legionella risk	Stagnant water, biofilm, warm water temperature (20-45°C range)	Regular biocide shock dosing, routine Legionella testing, maintain drift eliminators
Excessive drift/water loss	Damaged/missing drift eliminators, high wind	Inspect and replace eliminators, check fan speed/wind screens
Poor cooling (high cold water temp)	Fouled fill, low air flow, fan/belt issues, overloading beyond design TR	Clean/replace fill, check fan and belt condition, verify actual heat load vs design
Vibration/noise in fan	Belt misalignment, fan blade imbalance, bearing wear	Realign belts, balance fan blades, replace bearings per schedule
High make-up water consumption	Excessive blowdown, drift losses, leaks	Optimize COC via conductivity control, inspect basin/piping for leaks

8. Equipment Service Life

Equipment	Typical Service Life	Key Life-limiting Factors
FRP Tower Casing/Basin	15-20 years	UV degradation, chemical attack, structural fatigue
Fill Media (PVC film fill)	8-12 years	Scaling, fouling, UV/chemical degradation
Drift Eliminators	10-15 years	Physical damage, brittleness with age
Fan Blades (FRP/Aluminium)	10-15 years	Erosion, fatigue, imbalance-induced stress
Gearbox/Speed Reducer	15-20 years (oil change every 1-2 yrs)	Lubrication quality, misalignment
Motor	15-20 years	Moisture ingress, bearing wear, overload
Water Distribution Nozzles	8-10 years	Scaling/clogging, UV degradation (plastic types)
Circulation Pumps	8-10 years (seals/bearings 2-3 yrs)	Cavitation, seal wear, corrosion

9. Preventive Service & Maintenance Schedule

Frequency	Activity
Daily	Check basin water level and make-up valve; visual check for scale/algae; verify dosing pump operation; log conductivity/blowdown
Weekly	Test water chemistry (pH, conductivity, hardness, inhibitor residual); inspect fan belt tension; clean strainers
Monthly	Inspect fill media condition; check gearbox oil level; test biocide efficacy; inspect drift eliminators
Quarterly	Vibration analysis on fan/motor; corrosion coupon retrieval and analysis; clean basin sediment
Half-Yearly	Gearbox oil change; inspect/retorque fan blade fasteners; Legionella risk assessment
Annual	Complete shutdown, fill replacement/cleaning if fouled, full mechanical overhaul of fan/gearbox/motor, structural inspection of FRP/casing, repaint steel structure if applicable

10. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Cold water temperature higher than design	Fouled fill, reduced air flow, overloaded heat load	Clean fill, check fan RPM/belt, verify actual vs design heat load
High conductivity/TDS in basin	Blowdown valve stuck/undersized, high evaporation vs blowdown	Inspect and service blowdown valve, recalculate COC setpoint
White/brown scale deposits	Calcium carbonate/silica scaling	Acid clean (with care), review antiscalant program and COC
Unusual fan noise/vibration	Bearing wear, blade imbalance, loose fasteners	Shut down, inspect bearings/blades, balance and retorque
Water carryover beyond tower (drift)	Damaged drift eliminator, excessive fan speed	Inspect/replace eliminators, check fan speed against design

Rapid depletion of biocide/inhibitor stock

Under-dosing correction cycle, leaking dosing system, high blowdown loss

Check dosing pump calibration, inspect for tank/line leaks

11. Project Execution & Engineering Workflow

Phase	Key Deliverables / Activities
Heat Load & Site Study	Determine design heat rejection load (TR/kcal-hr), design wet bulb temperature, site layout and air recirculation study.
Basic Engineering	Tower type/cell count selection, fill type selection, pump/piping sizing, GA layout.
Detailed Engineering	Structural design (FRP/steel/concrete), fan and motor selection, water treatment system design, P&ID, electrical SLD.
Procurement	Tower cells, fans/motors/gearboxes, fill media, dosing systems, instrumentation.
Civil Works	Basin foundation, pump house, pipe racks/trenches.
Erection & MEI	Cell assembly, fan/motor installation, piping, dosing skid installation, instrumentation wiring.
Pre-commissioning	Water fill trial, mechanical run-in of fans, alignment checks, dosing pump calibration.
Commissioning	Full-load thermal performance trial against guaranteed range/approach at design conditions.
Performance Test	CTI (Cooling Technology Institute) or equivalent test code verification of thermal capability.
Handover & O&M	O&M manual, spares list, water treatment program handover, operator training.

12. Standards, Safety & Performance Guarantee

- Thermal performance testing typically per CTI ATC-105 or equivalent standard, verifying range, approach and water flow against design at actual ambient wet bulb during test.
- Legionella control per local guidelines (e.g., ACGIH/HSE L8-type good practice) – documented water treatment log and periodic microbiological testing mandatory for commercial/HVAC towers.
- Fire safety clearance required where FRP structure is used in industrial plants (combustibility rating of FRP resin to be verified).
- Guarantee typically covers range/approach at rated flow and design wet bulb, with liquidated damages for shortfall beyond an agreed tolerance (commonly $\pm 0.5^{\circ}\text{C}$ on approach).

Technical note: figures in this manual are general design guidelines. Final tower selection, chemical dosing and operating limits must be established from actual site meteorological and water quality data.